PIPELINE AUDIT AND HEALTH REPORT

Machining plant: parallel assembly + rework loop

Milling -> conveyor -> two parallel assembly cells (shortest-queue) -> inspection. Inspection ships 90%, scraps 2% and sends 8% back to assembly through a bypass rework lane.

Generated 2026-10-13 19:48 | simulated 2.0 h (warm-up 10 min) | seed 42 | 10 replications

HEALTH SCORE

65 /100

Grade D

At risk: significant constraints or fragility; fix the high-severity findings first.

Critical 0 | High 2 | Medium 3 | Low 1 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
243.3 units/h	489 units	33.8%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
22.0 units	306 s	asm_a

Executive summary

Primary bottleneck: 'asm_a' (92% busy, 8% down, index 105). 33.8% of arrivals are lost. Run what-if to measure which lever recovers the most throughput.

What to fix first

- High (arrivals): Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: 10% faster processing at 'asm_a' (+5.1% throughput).
- High (asm_a): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'asm_a' (+5.1% throughput).
- Medium (milling): Speed up or add capacity downstream of this station, or enlarge the buffer after it.
- Medium (asm_b): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.

Best measured improvement

Best lever: 10% faster processing at 'asm_a' -> +13.1 units/hour (5.1%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Flow	**arrivals**	33.8% of arrivals are lost because a buffer is full. Action: Add capacity at the constraint or enlarge the buffer in front of it. Measured best lever: 10% faster processing at 'asm_a' (+5.1% throughput).	-15
High	Capacity	**asm_a**	The bottleneck is 100% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'asm_a' (+5.1% throughput).	-8
Medium	Flow	**milling**	Blocked 27% of the time: finished parts cannot leave because the next buffer is full. Action: Speed up or add capacity downstream of this station, or enlarge the buffer after it.	-3
Medium	Reliability	**asm_b**	Down 27% of the time (4 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Medium	Reliability	**milling**	Every unit passes through this station, and it can fail. A single failure stops the whole plant. Action: Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.	-4
Low	Model	**run length**	The measured window (1.8 h) is only 22x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**asm_a**	Constraint of the line: 100% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'asm_a' (+5.1% throughput).	

Performance by node

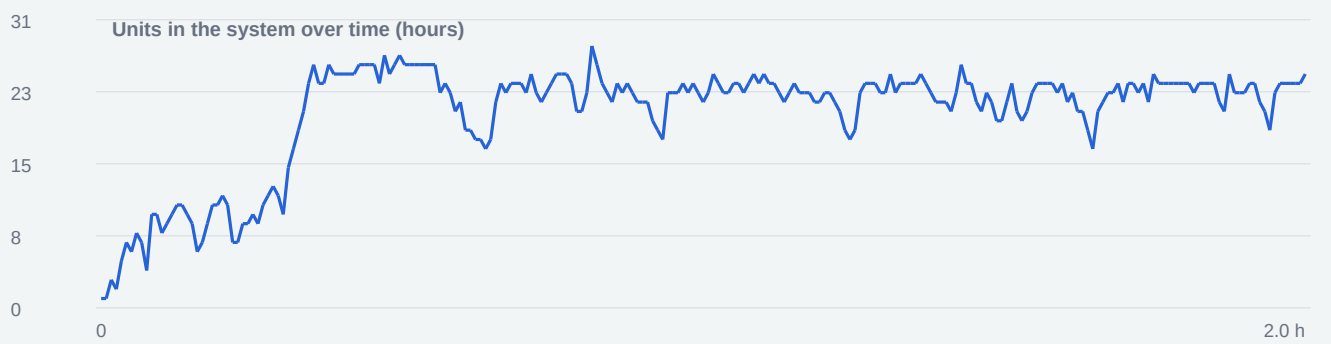
Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
arrivals	source	-	-	-	-	-
milling	machine		71.3	26.9	0.2	4.5 / 6
belt	delay		4.5	92.4	0.0	8.9 / 10
asm_a bottleneck	machine		91.7	0.0	8.0	3.9 / 5
asm_b	machine		72.4	0.1	26.8	4.1 / 6

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
inspect	machine		42.6	0.0	0.0	0.6 / 4
ship	sink	-	-	-	-	-
scrap	sink	-	-	-	-	-

Work in process over time



Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	259.42 units/h	+- 12.21	17.07	4.7%
Average WIP	20.25 units	+- 1.76	2.46	8.7%
Cycle time	274.19 s	+- 25.80	36.07	9.4%
Arrivals lost	26.43 %	+- 3.32	4.65	12.6%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
milling	60 min	5.0 min	92.3%	0.2%	0
asm_a	40 min	4.0 min	90.9%	8.0%	3
asm_b	40 min	4.0 min	90.9%	26.8%	4

Single points of failure

All units must pass through: milling, inspect. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 778 = completed 489 + scrapped 9 + lost 255 + in system 25
Little's Law	PASS	L = 22.04, lambda x W = 21.03 (gap 4.6%)
Run length	WARN	window = 22 x cycle time (30x or more recommended)
Warm-up	PASS	first 10 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 254.9 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
asm_a	10% faster processing	268.0	**+13.1**	+5.1%
asm_b	10% faster processing	267.8	**+12.9**	+5.0%
asm_b	20% shorter repairs	257.6	**+2.7**	+1.1%
milling	20% shorter repairs	256.8	**+1.9**	+0.7%
asm_a	20% shorter repairs	256.7	**+1.7**	+0.7%
milling	10% faster processing	256.4	**+1.5**	+0.6%
asm_b	+2 buffer slots	255.5	**+0.6**	+0.2%
asm_a	+2 buffer slots	255.3	**+0.4**	+0.2%
milling	+2 buffer slots	255.2	**+0.3**	+0.1%

Best lever: 10% faster processing at 'asm_a' -> +13.1 units/hour (5.1%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
arrivals	source	interarrival=exponential(10)
milling	machine	capacity=5, mtbf=3600, mttr=300, process=normal(10,1.50)
belt	delay	capacity=10, transit=6
asm_a	machine	capacity=3, mtbf=2400, mttr=240, process=normal(22,3)
asm_b	machine	capacity=3, mtbf=2400, mttr=240, process=normal(22,3)
inspect	machine	capacity=3, process=exponential(6)

NODE	TYPE	EFFECTIVE PARAMETERS
ship	sink	counts_as=completed
scrap	sink	counts_as=scrapped

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**arrivals**	33.8% of arrivals are lost because a buffer is full.	-15
High	**asm_a**	The bottleneck is 100% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**milling**	Blocked 27% of the time: finished parts cannot leave because the next buffer is full.	-3
Medium	**asm_b**	Down 27% of the time (4 failures).	-3
Medium	**milling**	Every unit passes through this station, and it can fail. A single failure stops the whole plant.	-4
Low	**run length**	The measured window (1.8 h) is only 22x the cycle time.	-2
	Total deductions		**-35**
	Health score		**65**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.